# Heterogeneous Belief-Driven Multi-Robot Search

**Multi-agent reinforcement learning for locating undocumented oil and gas wells with mixed aerial–ground robot teams**

Research report · Ali Pira · Oct 28, 2026

## Summary

This project builds a multi-agent reinforcement learning (MARL) system in which a mixed team of aerial and ground robots finds an unknown number of buried oil and gas wells from noisy sensor readings. Aerial robots screen wide areas with coarse sensors; ground robots confirm candidates with precise ones.

**The gap.** Belief-driven source search is studied almost only for single or identical agents. Multi-robot exploration treats targets as directly visible. Heterogeneous MARL has the policy machinery but has not been applied to belief-based search. No published system combines physically and sensor-heterogeneous robots, a fused Bayesian belief over an unknown number of sources, continuous control, and centralised training with decentralised execution (CTDE).

**The approach.** HBS (Heterogeneous Belief-driven Search) plans over a shared log-odds belief map. It rewards information gain measured as KL divergence, and only ground robots can confirm a well, by physical peak detection. A custom graph policy shares parameters within a robot type while passing messages across types, trained through a homogeneous-to-heterogeneous curriculum.

**Status (October 2026).** The literature review, gap analysis and architecture are complete. The simulation environment and four scripted baselines are built on VMAS and BenchMARL and checked by executable test contracts: 93 tests across two completed build stages, each contract validated by mutation testing. Learning experiments have not run yet, so no learned-policy results are claimed. Building and measuring changed the design in four ways:

- The intuitive intrinsic reward, entropy reduction, turned out to penalise discovering a source. KL information gain replaces it.
- A belief-threshold confirmation rule produced 3.2–3.8 false positives per true source. Physical confirmation brings this to 0.26–0.29 for a scripted team, at a recall cost (0.99 down to 0.84–0.88).
- No scripted policy lowers false positives further, and seeking information makes them worse. Every false positive is a ground robot passing a source off-centre; a robot that approached head-on would cut the rate from 0.30 to 0.04 per source. That gap is what learning has to win.
- Two library defaults would silently break heterogeneous training: separate robot types could not exchange graph messages, and timeouts were marked as terminal. Both are fixed by design and pinned by tests.

## Problem and motivation

Undocumented orphan wells are a large, funded search problem that is still solved with scripted surveys. The United States alone has hundreds of thousands of them, mostly without a recorded location, many leaking methane or contaminating groundwater (O'Malley et al., 2024). By 2021, states had already identified more than 129,000 documented orphaned wells. The Bipartisan Infrastructure Law set aside $4.7 billion to plug them (US DOI), though disbursement was paused in 2025 (E&E News).

**How wells are found today.** Steel well casing produces a magnetic anomaly. Aircraft carrying magnetometers fly fixed parallel lines, and ground crews then visit each candidate to verify it. Helicopter magnetometer surveys found 91–96% of the wells located by exhaustive ground search at two Wyoming fields (Hammack et al., 2023). The screening is good; the cost sits in the ground follow-up, and every false candidate is a wasted trip.

**Formulation.** I model the task as a decentralised partially observable Markov decision process (Dec-POMDP) with two robot types:

- **UAVs** are fast and see a wide footprint through a noisy sensor, but cannot confirm a well.
- **UGVs** (differential-drive ground robots) are slow, read a precise point sensor, and are the only robots that can confirm.

An unknown number K of sources sits in the field. Robots never observe the sources directly, only a noisy reading of the anomaly field at their position:

$$c(x) = \sum_{k=1}^{K} A_k \exp\left(-\frac{\|x - p_k\|^2}{2\sigma_k^2}\right), \qquad z_i = c(x_i) + \varepsilon_i$$

Amplitude A, width σ and the number of sources K are randomised per episode. The team must confirm as many sources as possible within a time budget while declaring as few false wells as possible. Because the sources are hidden, the robots act on a fused Bayesian belief over where the sources are, which makes this a belief-space planning problem rather than plain navigation.

## Related work and the research gap

I reviewed about 25 papers from 2023–2026 in four clusters. Each cluster solves part of the problem, and none covers all of it:

1. **Belief-driven source search** (gas and radiation localisation). It has the right task machinery: particle and grid filters, information-theoretic rewards, stopping rules. The agents are single or identical and usually chase one source. Examples are PC-DQN, DDPG-based source term estimation, RAD-TEAM, and learned cessation for source term estimation (AAAI 2025).
2. **Multi-robot informative path planning and learned exploration.** It brings budgets, communication limits and attention or graph policies that scale (Vashisth et al., 2024; MARVEL, ICRA 2025; privileged communication learning, 2024). Teams are homogeneous, and targets are visible objects rather than inferred field sources.

3. **Heterogeneous MARL.** It supplies per-agent parameters with graph communication (HetGPPO), diversity measurement and control (SND, DiCo), intrinsic rewards for sparse settings (CoHet) and staged training for a UAV–UGV pair (Chen and Xiao, 2023). It is validated on abstract coordination tasks with no belief and no confirmation step. The HeMAC benchmark reports that current algorithms still struggle with heterogeneity.
4. **Infrastructure.** VMAS, TorchRL and BenchMARL make the whole algorithm ladder runnable on one vectorised simulator.

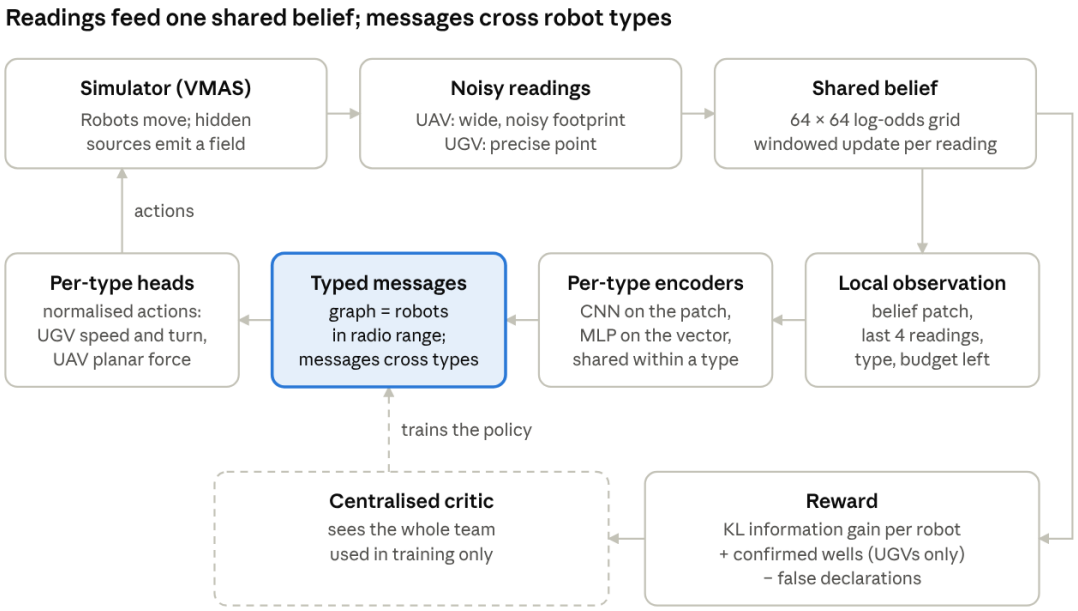
| Representative work | Multi-agent | Heterogeneous | Belief over sources | Unknown, multiple sources | Continuous control |
|---|---|---|---|---|---|
| PC-DQN, DDPG source term estimation (2022) | no | no | yes | no | DDPG only |
| RAD-TEAM (2023–24) | yes | no | yes | no | no |
| Scalable multi-robot IPP (2024–25) | yes | no | partial | visible targets | no |
| HetGPPO (2023) | yes | yes | no | no | yes |
| CoHet (2024) | yes | yes | no | no | yes |
| UAV–UGV staged search (2023) | 2 robots | yes | no | no | yes |
| **HBS (this work)** | **yes** | **yes** | **yes** | **yes** | **yes** |

**Gaps this project targets:**

- **G1 (core).** No system combines heterogeneous platforms and sensors, a fused belief over an unknown number of sources, continuous control and scalable CTDE.
- **G2.** Intrinsic rewards for heterogeneous teams are task-agnostic. The belief offers a task-grounded signal: information gain, which has to be measured as KL divergence (Section 5 shows why).
- **G3.** Search with an unknown number of sources needs the team to decide when it is done; multi-agent learned stopping is open.
- **G4.** Warm-starting typed policies from a homogeneous phase is common practice but has not been systematised or evaluated.
- **G5.** Robustness to sensor noise, communication dropout and team size has not been evaluated jointly for search.
- **G6.** Credit for a discovery spans robot types and time (the UAV screened, the UGV confirmed), and a scalar team return hides it.

## Proposed approach: HBS

HBS couples a Bayesian belief over where the wells are with a graph policy that lets each robot type specialise while still coordinating across types.




HBS decision step · execution loop above, training-only critic dashed

Each step runs the loop clockwise from the simulator. The highlighted block is the core contribution: message passing that crosses robot types while parameters stay shared within each type.

**Belief.** Each step, every reading updates a 64×64 log-odds grid. For a cell at distance d from the sensor, μ is the reading expected if the source were in that cell, and the update is a clipped Gaussian likelihood ratio against "no source here":

$$\ell_j \leftarrow \ell_j + \mathrm{clip}\left(\frac{z^2 - (z-\mu_j)^2}{2\sigma_n^2}\right), \qquad \mu_j = a\exp\left(-\frac{d_j^2}{2s^2}\right)$$

Only cells within the sensor's influence radius change, so the update is windowed: about 441 cells per reading instead of 4,096.

**Reward.** Each robot is rewarded for the information its own reading adds, measured as the KL divergence between the belief after and before:

$$r_i^{\text{info}} = \sum_j \mathrm{KL}\Big(\mathrm{Bern}(p_j^{\text{post}}) \,\|\, \mathrm{Bern}(p_j^{\text{prior}})\Big) \geq 0$$

The team shares a large reward for each confirmed well and a penalty for each false declaration. Small per-robot costs cover energy and collisions.

**Confirmation.** Only UGVs confirm. A UGV declares a well where its precise reading peaked along its path, as a ground crew walking a magnetometer line would. A declaration is a true positive if a real source lies within tolerance, otherwise a false positive. UAVs screen and UGVs confirm, so the heterogeneity is necessary to the task rather than decorative.

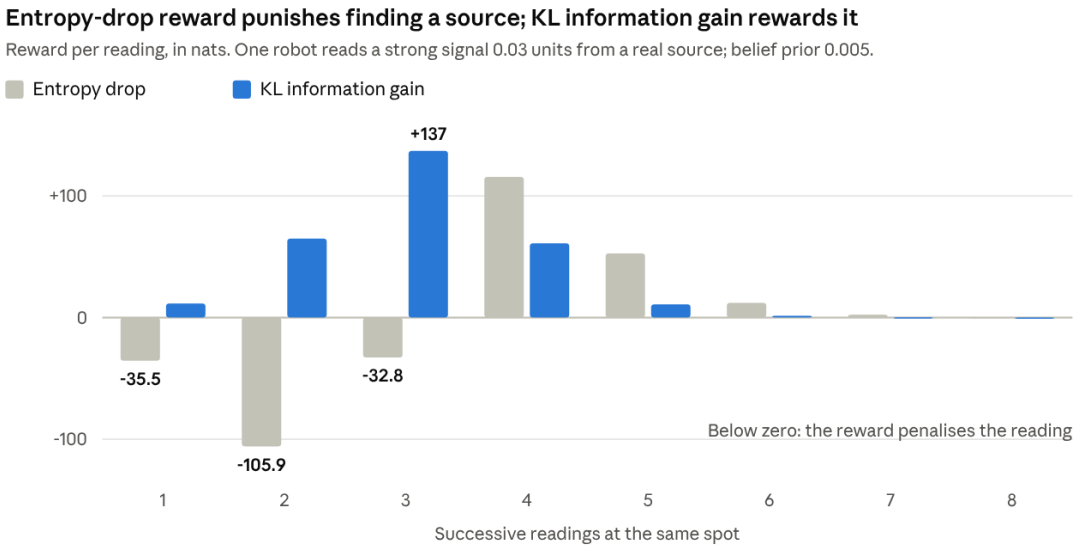
**Policy.** All robots form one team group, sharing one observation and action format with the robot type encoded in the observation. A custom graph model gives each type its own encoder for its local belief patch and readings. It builds the communication graph from positions within radio range and passes type-conditioned messages across types: parameters are shared within a type, information flows between types. Training is centralised (a critic sees the whole team); execution is decentralised.

**Curriculum and baselines.** Training runs from a homogeneous ground team to a mixed team, warm-starting the typed policies from the shared one (G4). Every learned method is compared with the same ladder on the same simulator:

| Rung | Method | What it isolates |
|---|---|---|
| B0 | Scripted lawnmower; lawnmower with homing; greedy information gain; infotaxis-style | Industry practice and classical search |
| B1–B2 | Independent DDPG (one agent, then several) | Learning without coordination |
| B3 | MADDPG | Off-policy centralised critic |
| B4–B5 | IPPO, MAPPO | On-policy, independent vs centralised critic |
| B6 | MAPPO with a graph network | Value of communication |
| HetGPPO | IPPO with a graph network and per-agent parameters | Published heterogeneous baseline |
| **HBS** | Typed graph model with belief, KL reward and curriculum | The contribution |

## Validation so far

Building the environment and baselines on the real stack (PyTorch 2.14, VMAS 1.5.2, BenchMARL 1.5.2, TorchRL 0.11.1) produced eleven measured findings. Each changed the design or the plan, and each is now a regression test or an acceptance gate. The most important concerns the reward itself:




Prototype measurement, October 2026 · one robot, eight readings at one real source

Under a sparse prior, positive evidence moves cells from p≈0.005 toward 0.5, which raises entropy. Entropy drop therefore punishes exactly the readings that discover a well. KL divergence between the belief after and before a reading is never negative and peaks at discovery.

| Finding | Measured | Design response |
|---|---|---|
| Entropy-drop reward penalises discovery | See chart | KL information gain |
| Belief-threshold confirmation over-declares | 3.2–3.8 false positives per true source: saturated cells tie, and `argmax` picks an arbitrary one | Physical confirmation at the peak of a UGV's reading: 0.26–0.29, at a recall cost |
| Graph messages cannot cross robot types | BenchMARL builds one model per agent group, and types land in separate groups by name | One team group; type encoded in the observation |
| Graph policy refuses any mixed team | Different physical action ranges per type create a heterogeneous action spec | All types act in $[-1, 1]^2$, scaled per type inside the simulator |
| Timeouts treated as terminal | The VMAS wrapper sets `terminated` at the horizon and never `truncated` | Remaining budget in the observation, making the horizon a true terminal (Pardo et al., 2018) |
| Ground robots run slow | Default drag delivers exactly 80% of the commanded speed | Drag set to zero; tracking within 2% |

| Finding | Measured | Design response |
|---|---|---|
| Belief update dominates compute | 63–72% of each simulation step at 300 parallel environments | Windowed update: 15–20%, 2.3× throughput |
| A CNN cannot precede a position-reading graph layer | Stock BenchMARL allows such a layer only first in a model chain | The custom HBS model owns the whole chain |
| Metrics kept counting after episodes ended | Batched simulation keeps stepping finished episodes, so early false-positive figures included those steps | Every metric frozen at each episode's end; all figures here are corrected |
| Scripted homing does not cut false positives | Every false positive is a pass 0.10–0.25 from a real source; homing scores 0.365 per source against the lawnmower's 0.301 | Homing kept as the recall and speed baseline; the false-positive rate is left to learning |
| Information-seeking passes sources off-centre | A greedy-KL frontier gathers 36% more information than the lawnmower ($p = 0.008$), finds no more sources and declares 29% more false ones | KL gain stays a small auxiliary reward (weight 0.05); confirmation carries the task |

**Scripted baselines.** Four scripted teams of four ground robots were measured on eight held-out seeds (100–107, never used for tuning), 200 parallel episodes each:

| Policy | Recall | False positives per source | All sources found | Median steps to all | KL information (nats) |
|---|---|---|---|---|---|
| Lawnmower, 0.15 lane spacing | 0.843 | 0.301 | 0.71 | 96.5 | 1,241 |
| Lawnmower with homing | 0.884 | 0.365 | 0.78 | 90.2 | 1,323 |
| Greedy-KL frontier | 0.845 | 0.388 | 0.71 | 72.8 | 1,692 |
| Infotaxis-style | 0.778 | 0.311 | 0.61 | 66.2 | 1,241 |
| Oracle (sees the true sources; diagnostic only) | 1.000 | 0.042 | 1.00 | 22.5 | 748 |

Every false positive of every scripted policy is a localisation miss: a pass 0.10–0.25 units from a strong source still peaks above the confirmation threshold. Homing raises recall by 4 points and finishes 6.5 steps sooner on all eight seeds (paired $p = 0.008$), but each extra pass near a source risks another miss. Information-seeking gathers 36% more information yet finds no more sources: the most uncertain cells sit on the rims of belief discs, so it passes sources off-centre. The oracle, which drives straight at the true sources, scores 0.04. False positives are therefore a skill of approach geometry, and the gap from about 0.3 to 0.04 is what the learned team has to win.

**Method.** Each build stage is specified as an executable test contract: 35 tests for the environment, 27 for the metrics and homing baseline, and 31 for the information baselines and cost bridge, 93 in all. I

checked each contract with mutation testing, reintroducing defects on purpose; every defect that changes behaviour was caught. The baseline numbers reproduce across machines. All eight learning rungs, including the HetGPPO configuration on a mixed team, train an iteration on the scenario; this checks the wiring, not learning.

## Why it matters beyond the benchmark

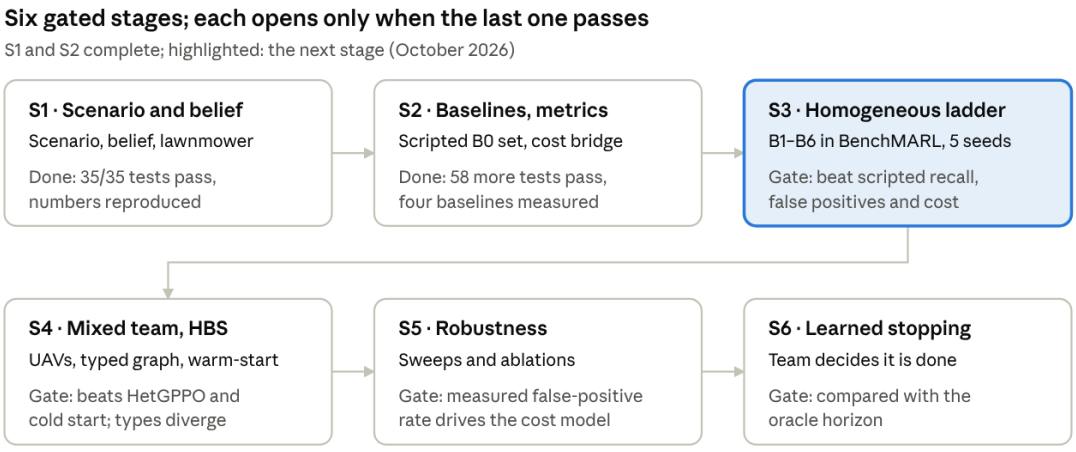
A cost model of a real survey campaign says the research should optimise false-positive discrimination, not coverage speed. I built a parametric model of a 500 km² search for 900 wells, costing staff, machines and time for three methods: ground crews, scripted UAV survey and a mixed robot team. The robot-team case takes its recall and false-positive rate from the simulation.

- **Confirmation is the bottleneck.** At every plausible false-positive rate, ground confirmation takes 26–37 days against 15.6 days of aerial screening. A faster aerial policy buys almost nothing.
- **Most of the saving comes from the team structure, not from learning.** The model projects about 46% savings for a screen-and-confirm robot team over scripted UAV survey with human ground checks. With the measured scripted robot team it already reaches $453 per well found, against $438 for the learned team I had assumed.
- **What learning can still win is false-positive rate.** The oracle reaches $350 per well, 23% below the scripted team, and almost all of that gap is false positives. Recall bought with false positives does not pay: homing finds 4 points more wells but raises the cost to $467 per well.

This is why the confirmation step is physical, why false declarations carry a penalty, why policies are ranked by cost per well found, and why false positives per true source is a headline metric beside recall. All cost figures are modelled, not field-measured.

## Research plan

The work runs in six stages. Each ships with an executable test contract and must pass a measured gate before the next begins.




Implementation roadmap · stages S1–S6 with their gates

S1 and S2 are complete. S3 starts next, and its gate is set by the scripted baselines: the best learned team must beat 0.884 recall, 0.301 false positives per source and $453 per well found on the held-out seeds. S4 carries the main contribution, and S5–S6 address gaps G5 and G3.

**Evaluation protocol.**

- **Primary metrics:** recall within the time budget, false positives per true source, time to first and to all confirmations, localisation error, and information gain over time.
- **Statistics:** at least five seeds per configuration, reported as interquartile mean with 95% bootstrap confidence intervals, never best-seed curves.
- **Heterogeneity:** System Neural Diversity (SND) to check that the robot types actually behave differently, plus a per-source log of which robot screened and which confirmed (G6).
- **Ablations:** without the graph, without the belief patch, without the KL reward, shared vs typed parameters, without warm-start, without physical confirmation.
- **Robustness (G5):** sensor-noise sweeps, communication dropout, training team size vs test team size, and source counts unseen in training.

**Main risks.**

- **Sparse confirmations stall learning.** Mitigated by the KL reward and the curriculum.
- **Robot types converge to the same behaviour.** Mitigated by making confirmation require a ground robot, and checked with SND.
- **Sim-to-real gap.** Mitigated by domain randomisation of noise, amplitude, source count and team size, and by keeping the 2D scope explicit.

## Open questions for a PhD

These are the questions I would pursue beyond this project's scope:

1. **Better beliefs for unknown source counts.** The per-cell grid assumes cells are independent, which smears each source into a disc and causes false declarations. A PHD filter or a sparse Gaussian-process field model would represent "how many sources, and where" directly. How much of the false-positive problem would a better belief remove before any learning?
2. **Credit across types and time.** A UGV's confirmation often depends on a UAV's earlier screening. Can difference rewards or a per-source sequential credit signal make that dependency learnable without hand-shaping?
3. **When does heterogeneity pay?** Recent theory characterises when diversity increases team reward. Does sensor heterogeneity (wide and noisy vs narrow and precise) satisfy those conditions, and can that predict the best team mix?
4. **Learned stopping as a team.** How should a decentralised team decide the field is fully surveyed when the number of sources is unknown, trading false completion against wasted time?
5. **From simulation to a real survey.** Which parts transfer to real magnetometer data and real flight rules, and what is the smallest field trial that would test the cost model's claims?

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